16 Solve the simultaneous equations

$$2x + 3y = 10$$
$$4x - y = -1$$

(1) $2x + 3y = 10$

(2) $4x - y = -1$

(2) × 3: $12x - 3y = -3$

1 mark
Eliminate y

Add(1): $14x = 7$

$$\begin{array}{ccc} 14x & = & 7 \\ \div 14 & & \div 14 \\ x & = & 0.5 \end{array}$$

Substitute: $4 \times 0.5 - y = -1$

$2 - y = -1$, so $y = 3$.

$x =$ .....

1 mark
Substitute x=0.5

$y =$ .....

1 mark
x=0.5,y=3

(Total for Question 16 is 3 marks)